

TUTORIAL - 5

Machine Learning

From Correlation to Clinical Prediction (MIMIC Case Study)

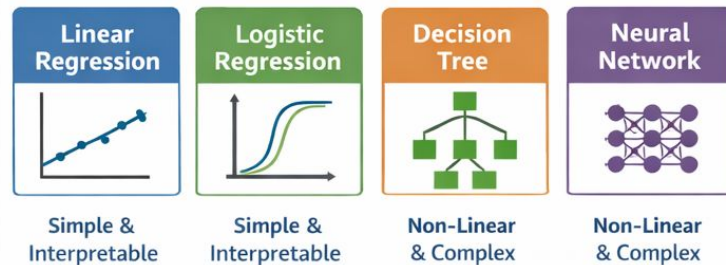
Learning Objectives

- Understand ML pipeline in healthcare
- Connect statistics → machine learning
- Perform feature engineering
- Evaluate clinical ML models
- Apply concepts to ICU (MIMIC) data

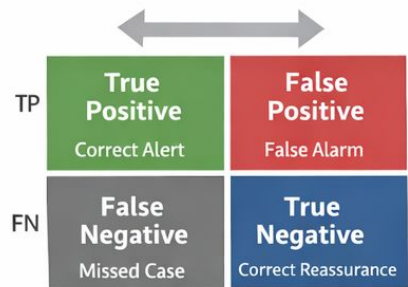
Feature Engineering Steps



Key ML Models in Healthcare

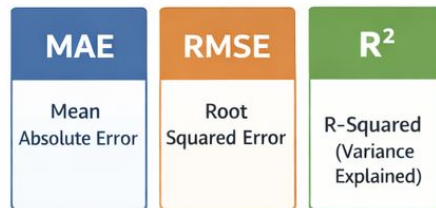


Classification Metrics Overview

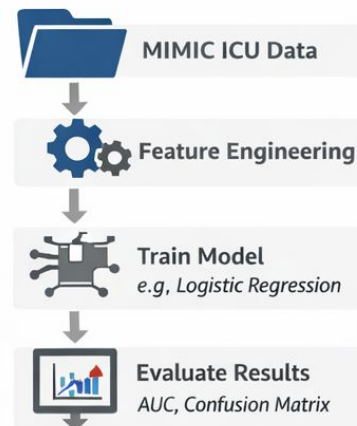


- Accuracy
- Precision
- Recall / F1 Score

Regression Model Evaluation



MIMIC Case Study Workflow



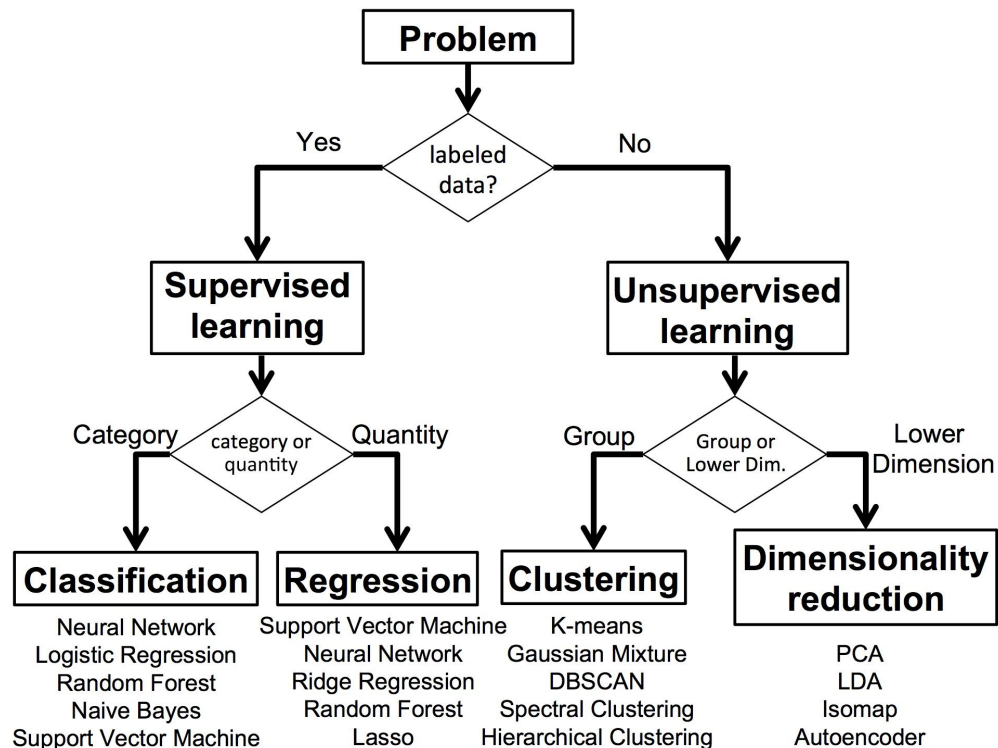
What is Machine Learning

Machine Learning = learning patterns from data to make predictions

Input: patient data (age, vitals, labs)

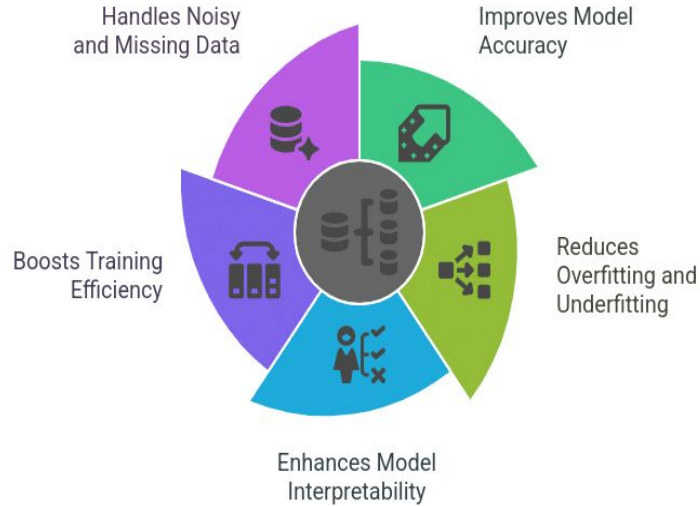
Output:

- Length of stay (LOS)
- Mortality (alive/dead)

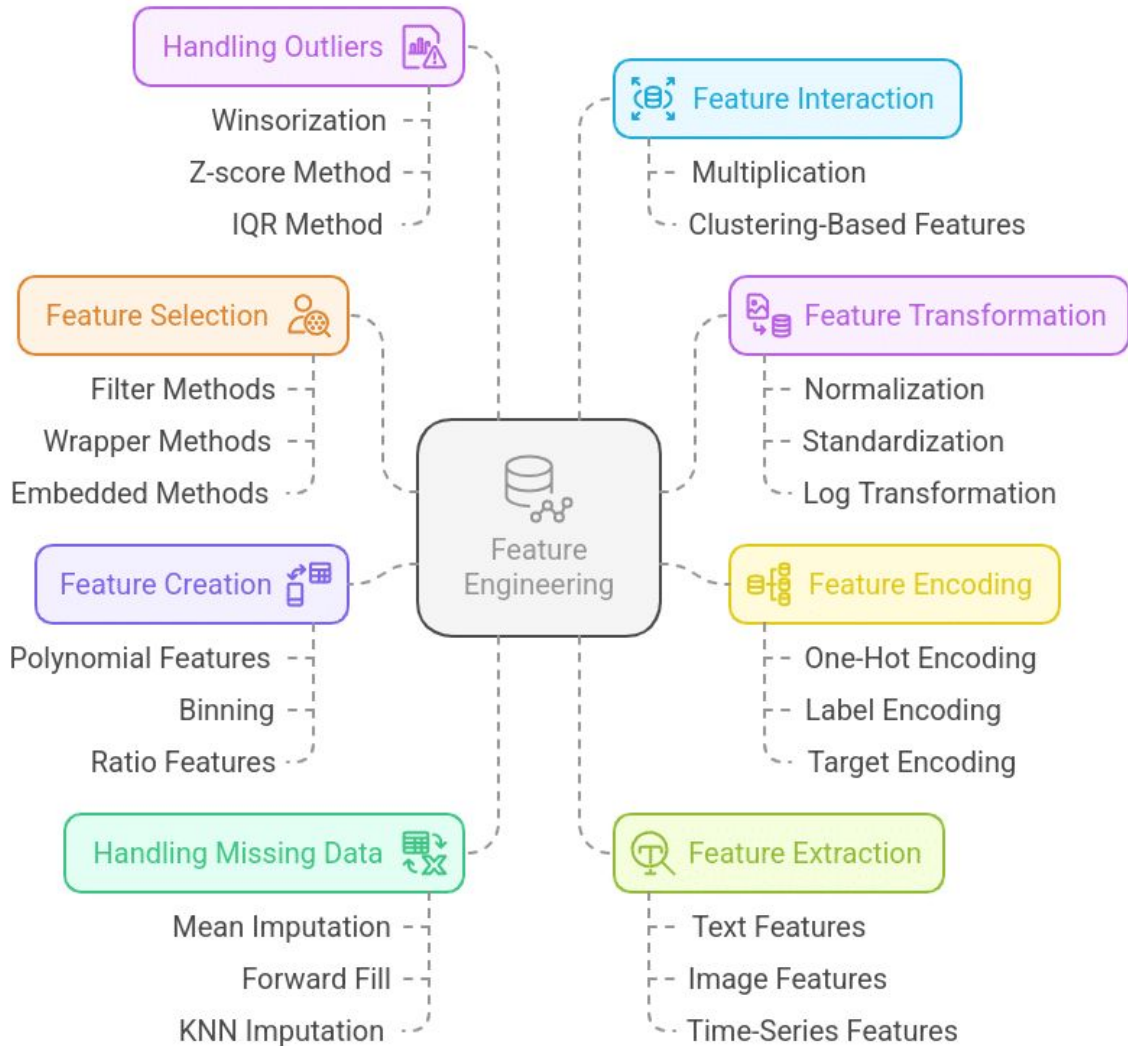


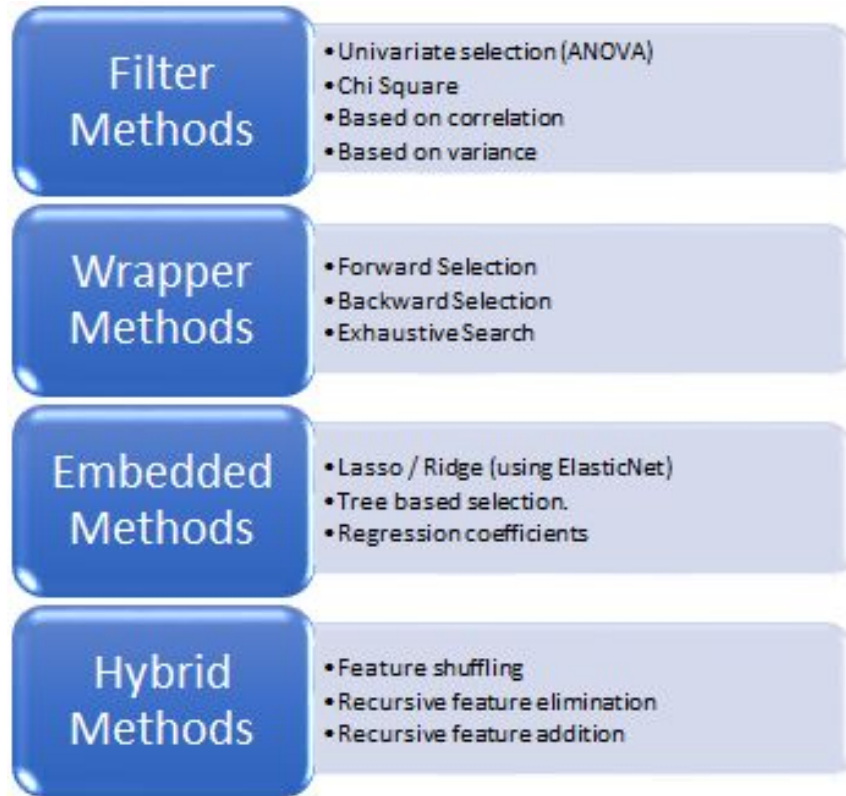
Data → Feature Engineering → Model → Evaluation → Interpretation

Feature Engineering



<https://www.mygreatlearning.com/blog/what-is-feature-engineering/>





Feature Selection Methods

1. Correlation-based

- Remove redundant variables
- Example: systolic BP & diastolic BP

2. Lasso (L1 regularization)

- Automatically removes unimportant features

3. Domain knowledge

- Clinically important variables (SOFA-like features)

Feature selection reduces noise and improves model performance.

Linear Regression

- Relationship Between Variables, Is a Linear Function
- Predicts continuous outcome
- Predict ICU length of stay
- $LOS = b_0 + b_1 \times \text{feature}$

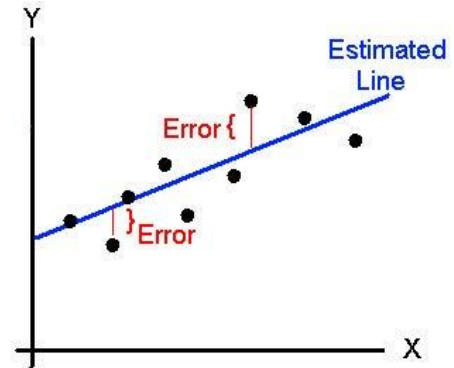
Estimated (or predicted) Y value for observation i

Estimate of the regression intercept

Estimate of the regression slope

Value of X for observation i

$$\hat{Y}_i = b_0 + b_1 X_i$$



Logistic Regression

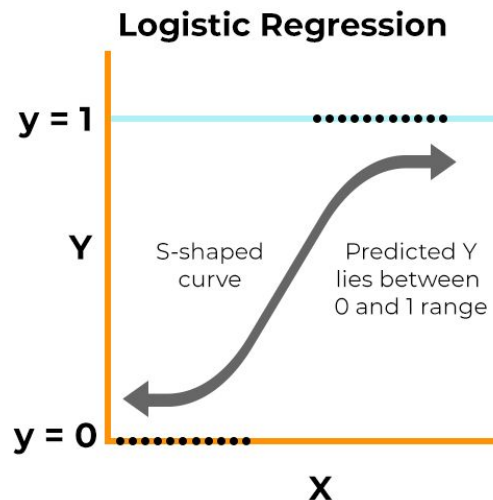
- A supervised machine learning algorithm used for binary classification (e.g., yes/no, true/false) by predicting the probability of an outcome.
- It fits an S-shaped curve (sigmoid function) to map linear inputs into probabilities between 0 and 1.
- The model calculates the best-fitting coefficients using maximum likelihood estimation.

a.k.a. **Log Odds**
or **Logit**

$$\log\left(\frac{P}{1-P}\right)$$

Intercept

$$= \beta_0 + \beta_1 X$$



Regression Metrics

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

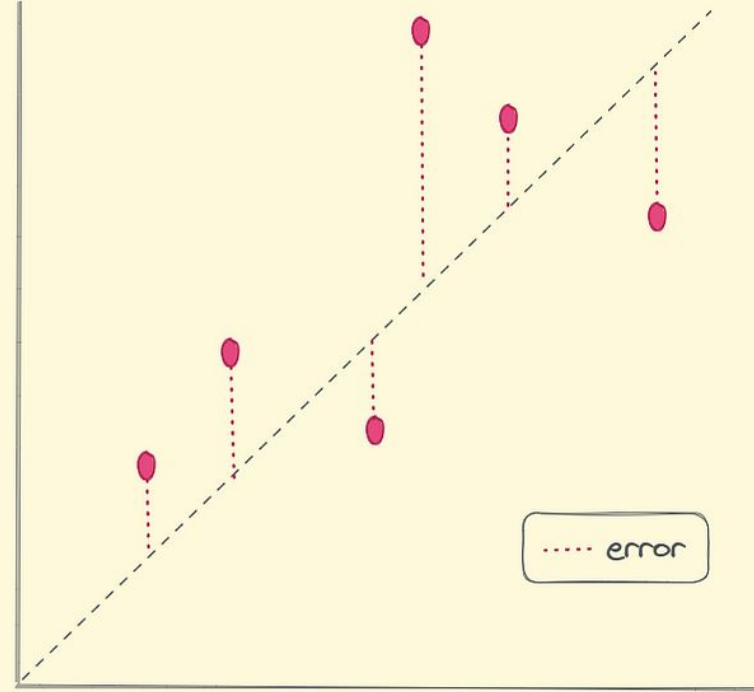
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Predicted Value



Actual Value

- **Mean Absolute Error (MAE):** Calculates the average absolute difference between actual and predicted values
Usage: Easy to interpret because it uses the same units as the target variable. Robust to outliers.
- **Mean Squared Error (MSE):** Calculates the average squared difference between actual and predicted values.
Usage: Heavily penalizes large errors, making it sensitive to outliers.
- **Root Mean Squared Error (RMSE):** The square root of MSE.
Usage: Like MSE, it penalizes outliers, but returns the metric to the original target units, making it more interpretable.
- **R-Squared (Coefficient of Determination):** Represents the proportion of variance in the dependent variable explained by the independent variables.
Usage: Ranges generally from 0 to 1, with 1 indicating a perfect fit.
- **Adjusted R-Squared:** A modified version of R-Squared that adjusts for the number of predictors, penalizing the addition of irrelevant variables.

Drawbacks in Biomedical ML

- **Data Quality and "Garbage In, Garbage Out:** inaccurate data, missing data, variability in measurement.
- **Dataset Characteristics:** limited sample size, class imbalance.
- **Model Design and Statistical Methods:** Data leakage, overfitting ignoring non linearities, improper evaluation metrics.
- **Clinical Implementation and Generalization:** lack of external validation, generalization failure, lack of interoperability.